CONFIDENTIAL

UTM UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

COURSE CODE	**:**	**SECD2613**
COURSE NAME	**:**	**SYSTEM ANALYSIS AND DESIGN**
YEAR / PROGRAM	**:**	**1 / 2 (SECB/SECJ/SECR/SECV/SECP)**
TIME	**:**	**8:00 P.M. – 10:00 P.M.**
DATE	**:**	**15 DECEMBER 2022**
PLACE	**:**	**BK2 to BK7 N28 and DEWAN KEJORA N28a**

INSTRUCTIONS TO CANDIDATES:

1. Fill in your particulars in the question booklet and answering booklet:

Name	
Passport No.	
Matric No.	
Year / Program	
Lecturer	

2. Read instructions given for each question carefully.

3. Answer **ALL** questions in this question paper.

4. This paper consists of three sections.
   Section A: **10 MULTIPLE CHOICE** questions (10 Marks).
   Section B: **4 STRUCTURED** questions (40 Marks).
   Section C: **1 CASE STUDY** question (30 Marks).
   TOTAL MARKS is **80 Marks**

THIS PAPER CONSISTS OF 10 (TEN) PRINTED PAGES INCLUDING THIS PAGE.

SECTION A : MULTIPLE CHOICE QUESTIONS
10 MARKS (EACH QUESTION 1 MARK)

INSTRUCTION : ANSWER ALL QUESTIONS.
Please write your answer (A/B/C/D) in the answer sheets.

Question 1
Which of the following is a **FALSE** statement regarding Upper CASE Tools?

A. Create and modify the system design
B. Generate computer source code
C. Help in modelling organizational requirements and defining system boundaries
D. Support prototyping of screen and report designs

Question 2
"Ecommerce uses the web to perform business activities."

Based on the above statement, which of these is **NOT** a benefit of using the Web?

A. Increasing awareness of the availability of the service, product, industry, person, or group
B. Improving the usefulness and usability of interface design
C. Creating a local system
D. 24-hour access for users

Question 3
"AC website is an online business which sells women's clothes. The owner of the website received many negative comments by the customers based on their experience while interacting with the website. Therefore, he decided to hire a systems analyst to improve AC ecommerce website. In a specific meeting, the owner asked his team to discuss the essential characteristics that the systems analyst should possess."

Based on the above scenario, which of these characteristics is the **MOST IMPORTANT** to a systems analyst?

A. Programmer
B. Project manager
C. Problem solver
D. Strong personal and professional ethics

Question 4
A functional requirement will describe a particular behavior of a system when certain conditions are met, for example:
"Send email when a new customer signs up".

Based on the above situation, during which phase of the System Development Life Cycle (SDLC) should the analyst identify the functional requirements of the new system?

A. Determining the stakeholders needs
B. Analyzing system needs
C. Designing the system
D. Implementing and evaluating the system

Question 5
Which of the following is **NOT** a merit to be considered as resource during determining feasibility?

A. Tactical
B. Operational
C. Technical
D. Economical

Question 6
In a PERT diagram, what is the **LONGEST PATH** called?

A. Slack time
B. Precedent activity path
C. Maximum path time
D. Critical path

Question 7
Which of the following is the **MOST** related aspect to the intangible cost?

A. It is quite impossible to estimate the exact projection amount of intangible cost for a proposed system
B. Development and production costs are the two intangible costs need to be estimated
C. Operational cost during system maintenance is included as intangible cost
D. Any quantified dollar amount that can be projected during project planning

Question 8
Which of the following answers is **NOT** an interactive method during information gathering?

A. Interview
B. Questionnaire
C. Observation
D. Joint Application Design

Question 9
Which is **NOT** considered as a complex random sample?

A. Systematic sampling
B. Cluster sampling
C. Purposive sampling
D. Stratified sampling

Question 10
Which of the statements below **BEST** demonstrate the pyramid structure?

A. Start with specific questions and end with general questions.
B. Start with general questions and end with specific questions.
C. Start with specific questions, move toward general questions and end with specific questions.
D. Start with general questions, move toward specific questions and end with general questions.

PART B	:	**STRUCTURED QUESTIONS** **40 MARKS (EACH QUESTION 10 MARKS)**
INSTRUCTION	:	**ANSWER ALL QUESTIONS.** **Please write your answer in the answer sheets.**

QUESTION 1

TESCO

TESCO is a world-class retail company and every day more than 11 million people visit TESCO stores. TESCO has got one of the most sophisticated retail supply chains in the world which makes it possible for all the guests to get whatever they need for the last 10 years. TESCO is working in more than 12 countries of the world. TESCO is the leading Superstore Company in the UK with more than 2000 stores in the UK. It specializes in many items including food, drinks, clothing, electronics, financial services, home appliances, health care, insurance, dental care, music, and telecommunication services, etc. TESCO is also one of the most successful online grocery shopping services in the world which has more than 750000 active customers across the world. Tesco is a very big organization comprising a variety of information systems, these systems assist the company in operating at levels ranging from the operational level to the executive. It is the structure of these systems which allows the business to run effectively, to ensure that the correct decisions are made and reports are compiled for effective management.

a. TESCO uses many Transaction Processing Systems (TPS) in its operation. The systems are responsible for serving the operational level of an organization. Name **THREE** TPS that are possibly to be used in TESCO.
   - Point of Sales (POS)

   (3 marks)

b. Management information system (MIS) is a computer-based system that can provide TESCO managers with essential tools for management. Explain **ONE** example of MIS that TESCO managers can use for effective management.

   (3 marks)

Part B

Question 1

(a) 1. Payroll TPS
2. Inventory Management
3. Point of Sale (POS)

(b) One example of MIS is a Sales Reporting MIS. This system collect data from TPS and generate summarized report for manager. For example, it can show sales trend and highlight underperforming items. Manager can also use this to make decision more efficiently.

(c) They can ensure the system meets business needs and operates efficiently. It also gather requirement accurately to ensure the system solves the right problem.
Other than that, it can improved decision-making as the system provides accurate and timely information.

c. How can a good system analysis and design help in the success of the development of any Information Systems in TESCO?

(4 marks)

QUESTION 2

UptownMY is an established shopping mall in Malaysia. In order to provide a unique shopping experience among their customers, the top management has decided to develop an e-commerce system embedded with augmented reality technology. As the system analyst, you need to conduct the economic feasibility study to ensure the successfulness of this project.

Before proceeding with the development of the system, you need to assess the economic feasibility based on the information given by the company below.

Development Costs	
Hardware	$20,000
Software	$10,000
Production Costs	
Supplies	$5,000 per year
Upgrades	$30,000 per year
Benefits	
Improve Customer Service	$2,000 per month
Increase productivity	$25,000 per year
Growth savings	$10,000 per year
Assumptions	
Discount rate	0.15
Sensitivity factor (cost)	0.5
Sensitivity factor (benefit)	0.9
Annual increment (cost)	0.05
Annual increment (benefit)	0.15

a. Calculate the cost-benefit estimation using the Present Value (PV) analysis for **THREE** (3) years. **(8 Marks)**

b. What is the Profitability Index (PI) value for this PV analysis? What is your recommendation based on the PI value and justify it. **(2 Marks)**

*Note: Answer the questions in **APPENDIX 1** provided at the end of the question booklet.*

APPENDIX 1 (QUESTION 2: CBA)

NAME :
MATRIC NO. :
SECTION :
NOTES : PLEASE DETACH THIS ANSWER SHEET AND ATTACHED IT WITH THE ANSWER BOOKLET. *Note: PV = Payment * (1/(1+C)^n)*

Item	Year 0	Year 1	Year 2	Year 3
Development Cost				
Hardware	10000			
Software	5000			
Total	15000			
Production Cost				
Supplies		2500	2625	2756.25
Upgrades		15000	15750	16537.50
Total		17500	18375	19293.75
Present Value (PV)		15217.39	13894.14	12685.95
Accumulated Costs		30217.39	44111.53	56797.48
Benefits				
Improve customer service		21600	24840	28566
increase productivity		22500	25875	29756.25
growth savings		9000	10350	11902.50
Total		53100	61065	70224.75
Present Value (PV)		46173.91	46173.91	46173.91
Accumulated Benefits		46173.91	92347.82	138521.73
Gain or Loss		15956.52	48236.29	81724.25
Profitability Index (PI)	5.45			
Justification	it is a good investment because of its index is more than one.			

QUESTION 3

A transportation company known as EZTransport wants to do an analysis of what to improve for their ineffective management of goods transportation. The company has branches and employees in 5 different cities. EZTransport has the culture of having a company retreat for three to four days for a strategic planning and team building among the multiple level of employees from all its different branches. Your team are given the task to gather information from EZTransport.

a. List **TWO** strong reasons why you want to propose JAD in place of personal organizational interviews for EZTransport. **(2 marks)**

b. List of **TWO** most important person that should be present in the JAD session. **(2 marks)**

During your visits to the company, you see there are lots of hardcopy of documents are available in the company's file room. You see that you there is a potential for you to understand the business in different perspective.

c. What is the suitable unobtrusive technique for reviewing the documents. Justify to the manager of why the technique is suitable to enable you the access for the documents in the file room. **(3 marks)**

d. List **THREE** types of documents that you need to borrow from the company's file room. **(3 marks)**

a) i) Multiple branches and employees in 5 different countries
- JAD brings all the stakeholder together in one session, saving time and travel cost to conducting individual interview in 5 cities.

ii) Faster decision making
- JAD allow all participant from different level to discuss and agree on requirement.

b) - Session leader - guide the JAD session
- Executive Sponsor - the ultimate authority to make decision about the project.

c) Suitable unobtrusive technique is using sampling
Document review is unobtrusive technique, so it does not distrupt employee's daily work. It's allow the analyst to examine existing document such as form, reports and record understand current processes objectively. Since the document already in file room, so reviewing them does not required employee time because it provides factual, historical evidence of how the business how it actual operate.

d) i) Transportation request form - show current process how transportation order are placed
ii) Delivery / Shipment record - show how movement is tracked between branches
iii) Management report - show performance metric and management decision-making process.

QUESTION 4

Figure 1 shows the DFD for Payroll System.

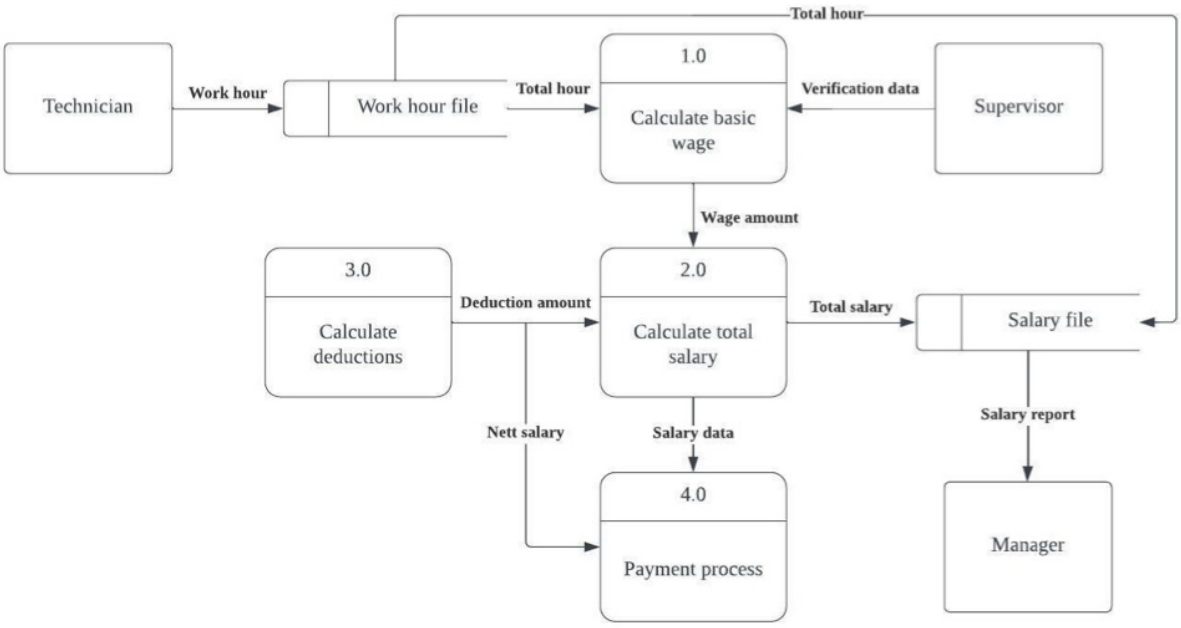


Figure 1: Payroll System DFD

a. What is the DFD level for Figure 1? **(1 mark)**
b. Identify all errors in the diagram **(4 marks)**
c. Based on your Question (b) answer, draw the correct DFD **(5 marks)**

SECTION C : CASE STUDY
30 MARKS

INSTRUCTION : ANSWER ALL QUESTIONS.
Please write your answer in the answer sheets.

You are the Project Manager for a software house company called Dream SW Home. One of the projects assigned to you is the Training Information Management System (TIMS) for a local training centre. The project is expected to be done in 6 months and the planning is shown in Gantt Chart (Figure 2) below:

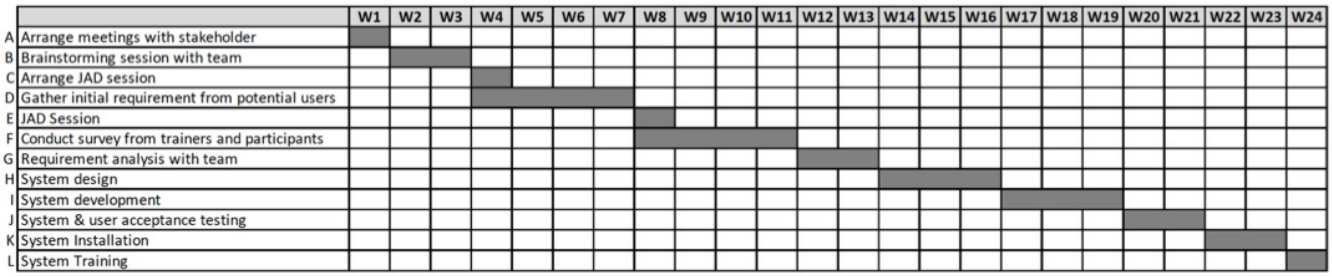


Figure 2 TIMS Gantt Chart

QUESTION 4

Figure 1 shows the DFD for Payroll System.

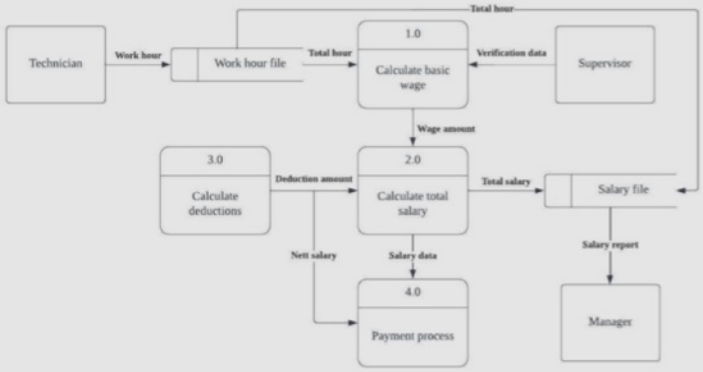


Figure 1: Payroll System DFD

a. What is the DFD level for Figure 1? **(1 mark)**
b. Identify all errors in the diagram **(4 marks)**
c. Based on your Question (b) answer, draw the correct DFD **(5 marks)**

a) diagram 0

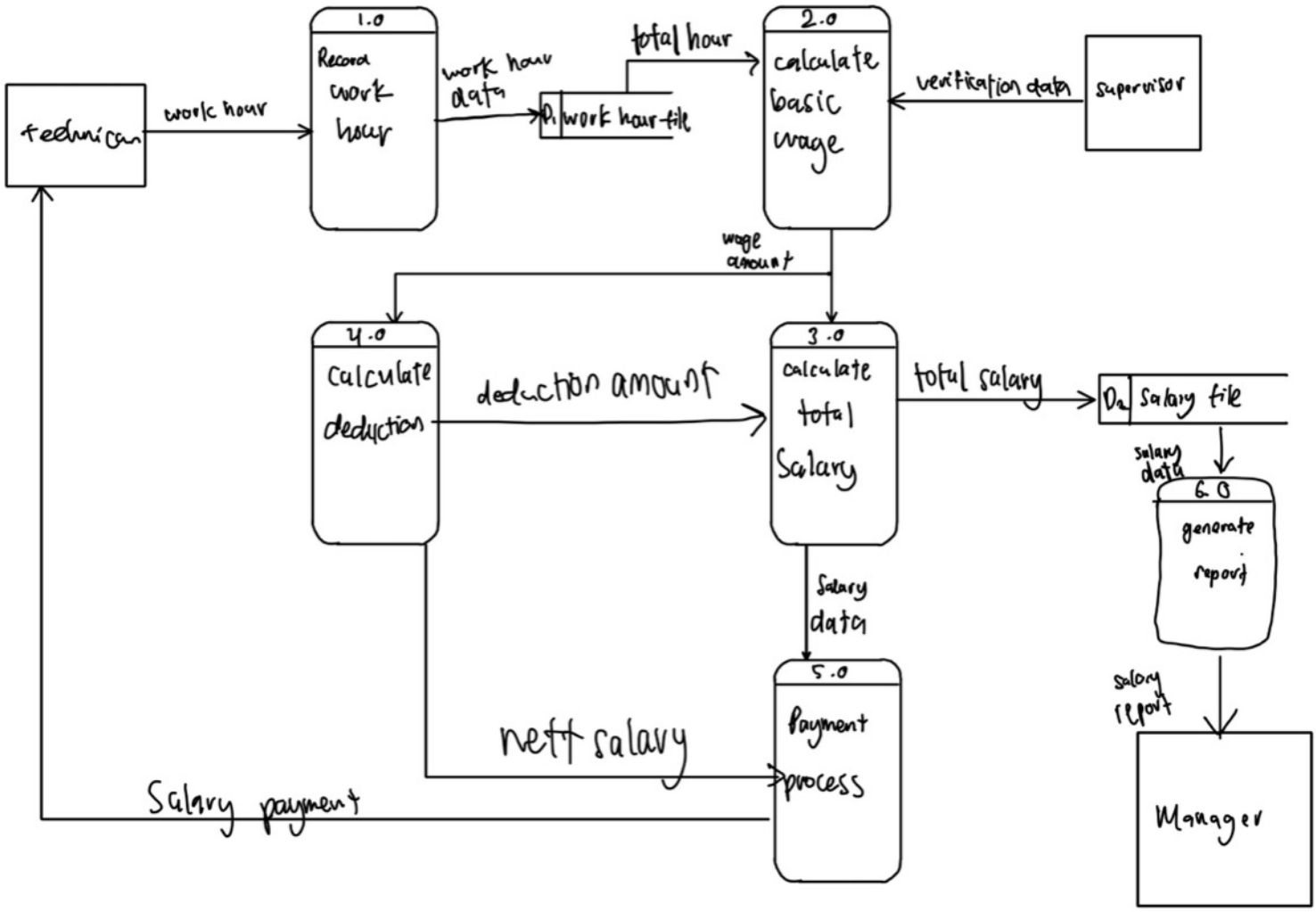
1) External entity (technican) connected directly to data store (work hour file). Data can't move from entity to data store

2) process (calculate deduction) has two output but no input. A process cannot create data out of nothing

3) process (payment process) has two input but no output. Every process must have at least 1 input and 1 output.

4) Data store (work hour file) to data store (salary file)
Data cannot move directly from one data store to another data store. Data must moved between store by a process.

c)



Part 1: Project Planning

After analysing the Gantt Chart, your team needs to plan and prepare all necessary materials before the project kick starts in two weeks' time.

a. Based on the Gantt Chart, create an equivalent PERT Chart to have a better view of the flow for each task. **[8 marks]**

b. From (a), list all the possible paths and its duration. **[4 marks]**

c. Identify the critical path from (b). **[1 mark]**

d. While preparing for the JAD session (Task C) in week 4, your team member informed you that the JAD session (Task E) is delayed due to unavailability of the Session Leader. Thus, the JAD session can only be done in week 10. How will this affect the duration of the project time? **[2 marks]**
There is **NO** effect to the project duration because Task G is dependent to Task E and Task F (have longer duration than Task E)

Part 2: Analysis Design

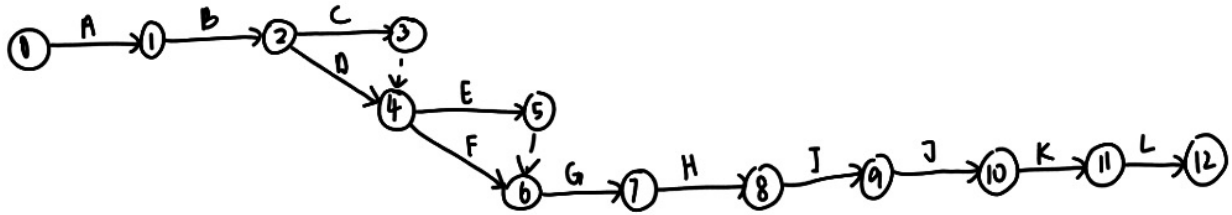
During the Requirement Analysis (Task G), your team summarized the current process of training management (as-is) as below:

> Currently, more than 1,000 participants are registered with the training centre, where they provide more than 20 courses to the participants. They also employ a number of trainers to conduct the training, and a few staff that handles the registration and management of the training.
>
> Every year, the training centre would plan the training calendar for all of their training. This will include the training content, expected number of participants, costs and duration of the training. Once this is done, the staff will give the schedule (training name, date and time) to their trainers so they would choose which training courses that they would like to teach.
>
> After the trainers have confirmed the training dates that they have chosen, the staff will give the trainers access to prepare the training material. The access to the material will open until the end of the training session. This includes all the notes and activities for the training.
>
> The details of the training will be advertised in all media platforms and training request forms will also be distributed. The advertisement includes the training name, dates, fee and account number for the payment. The staff will also email the training info to the past participants, and they will get 10% discounts of the training fees.
>
> New participants will need to register and provide their details such as name, date of birth, address, phone number and credit card number. After registration, the new participants will be given a unique ID. Payment details (training fees, account details

Case Study

Part 1

(a)



(b) Path 1: A-B-D-F-G-H-I-J-K-L

duration = 24 weeks

Path 2: A-B-C-F-G-H-I-J-K-L

duration: 21 weeks

Path 3: A-B-D-E-G-H-I-J-K-L

duration = 21 weeks

Path 4: A-B-C-E-G-H-I-J-K-L

duration = 18 weeks

(c) Path 1

(d) It will not effect the project duration because G depends on both E and F. F takes longer than E. So, by the time F finishes, E is already done even it have a delay.

and due date for payment) will also be given.

To register and pay for the training, the participants would just need to contact the staff and let them know which course they would want to attend. The training fee would be automatically charged to their account (details of the participant's credit card is already saved by the training centre staff). And once the payment is confirmed, their name will be updated in the training participants list.

On the day of the training, the staff will pass the list of participants to the assigned trainer. At the end of the training, the trainer will update the participants' attendance to the staff. The staff will then print out the certificate and pass it to the participants before they leave after the training. The participants' attended course information will also be updated in the participant's file for future reference.

Based on the case study, your team was asked to draw the as-is Data Flow Diagram (DFD) to have a better understanding of the current process flow. As the Project Manager, you have designed the Context Diagram as below:

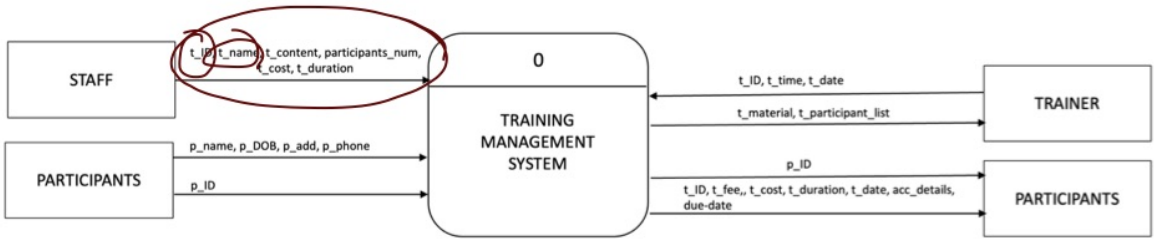


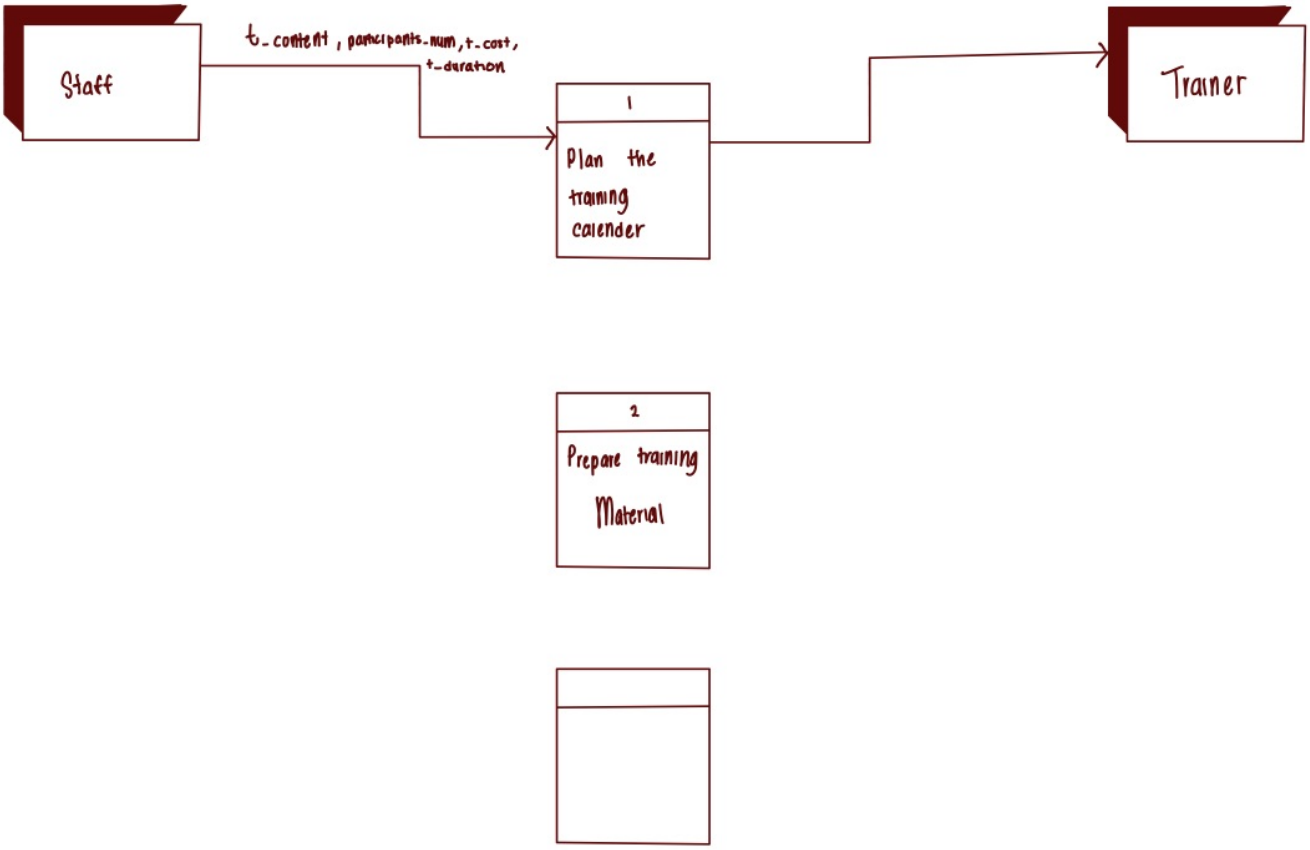
Figure 3: Context Diagram

a. You have identified six main processes for the Training Management System based on the discussion with your team members. The processes are: ***Plan the training calendar***, ***Prepare Training Material***, ***Advertise the training***, ***Register new participants, Register and Pay for Training*** and ***Produce certificate***. Develop a Data Flow Diagram (DFD) Level 0 for the current Training Management System (as-is). Your design must be based on the Context Diagram in Figure xx and the identified processes.

[10 marks]

b. One of the main processes in (a) is ***Register and Pay for Training***. Draw the child diagram (DFD Level 1) for this process. **[5 marks]**

a)


Staff
t_content , participants_num, t_cost, t_duration
1
Plan the training calender
Trainer
2
Prepare training Material